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Wright

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(54) **PORTABLE LIQUID COVER SYSTEM**

(56) **References Cited**

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A46B 15/00 (2006.01)

B05C 17/03 (2006.01)

(57) **ABSTRACT**

(52) **U.S. Cl.**

CPC **A46B 11/063** (2013.01); **A46B 15/0095** (2013.01); **B05C 17/0333** (2013.01); **A46B 11/06** (2013.01); **A46B 2200/202** (2013.01)

A liquid cover system comprising a vessel, a brush, and a hose, which efficiently applies liquid cover to a surface. The system includes an upper chamber and a lower chamber separated by a flexible diaphragm, which are pressurized by a handle connected to a pump body to force liquid cover through an outlet and the hose to the brush. The brush includes a pliable tube held at a slight angle towards one edge of bristles, to efficiently apply liquid cover to trailing edge bristles. The brush is secured to a handle, which includes a lever for controlling paint flow and a threaded outlet, and a sliding dovetail connection provided by a top joint and a bottom joint.

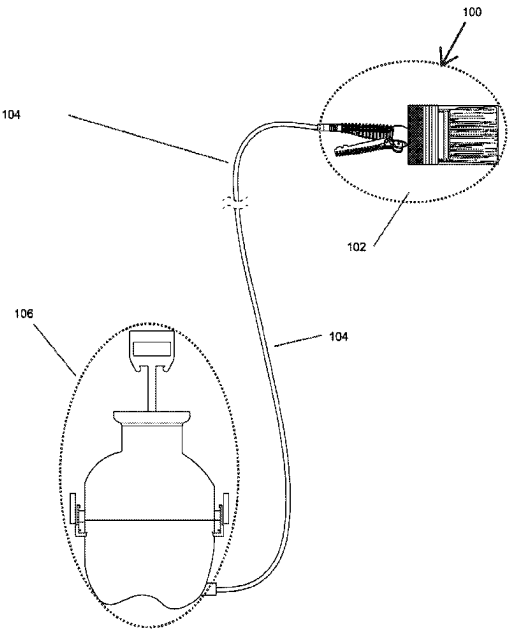
(58) **Field of Classification Search**

CPC A46B 11/063; A46B 15/0095; A46B 2200/202; A46B 11/06; A46B 11/00; A46B 11/0072; A46B 2200/20; B05C 17/0333; B05C 17/02; B05C 17/03; B05C 17/0357

USPC ... 401/152, 156, 157, 289, 188 R, 270, 286, 401/287

See application file for complete search history.

2 Claims, 5 Drawing Sheets



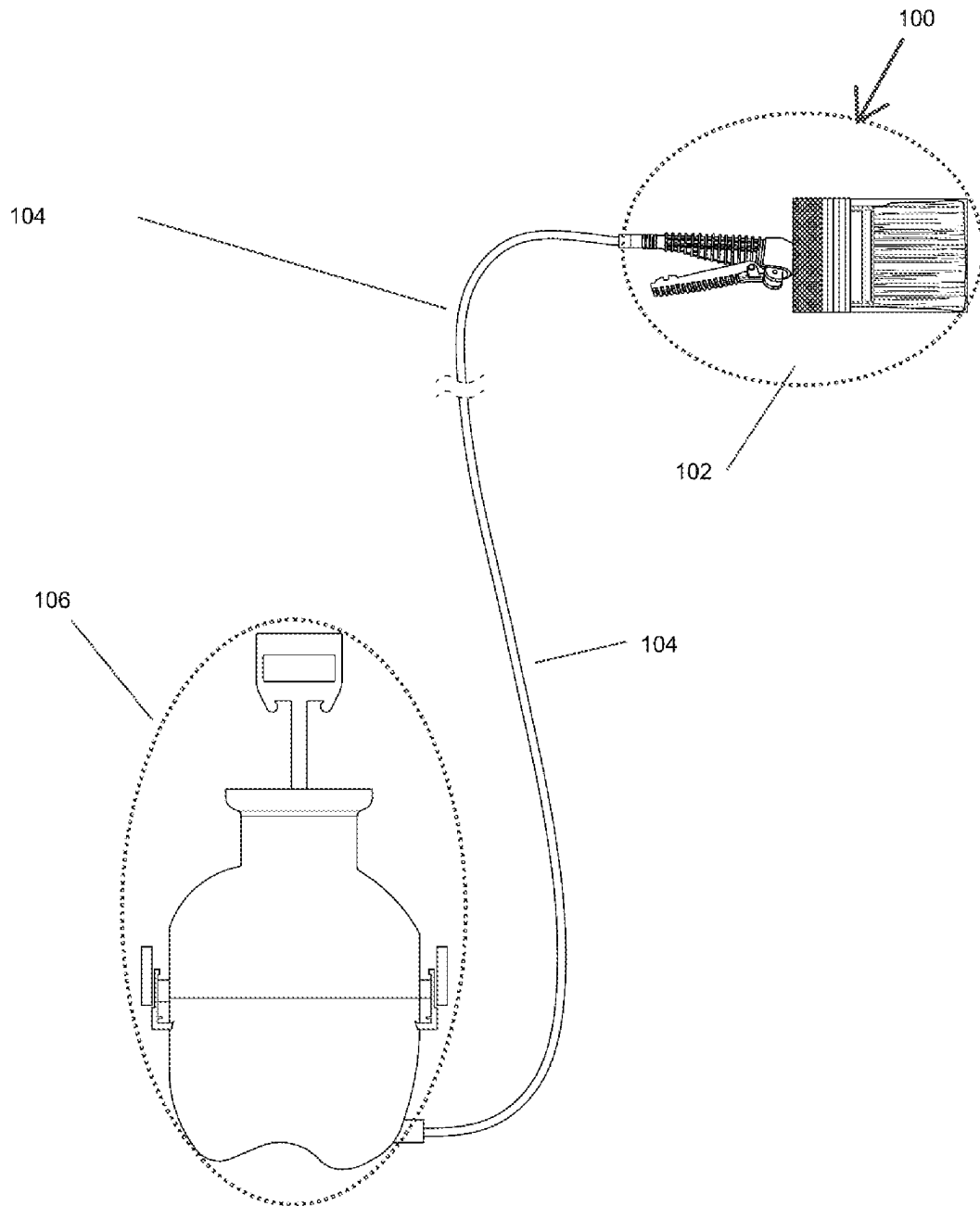


FIG. 1

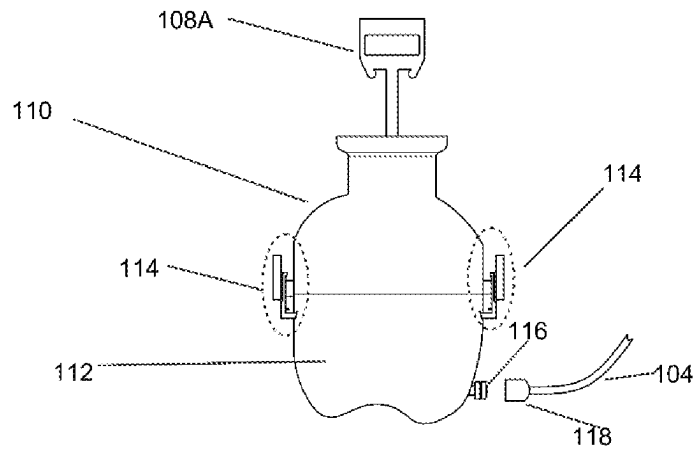


FIG. 2A

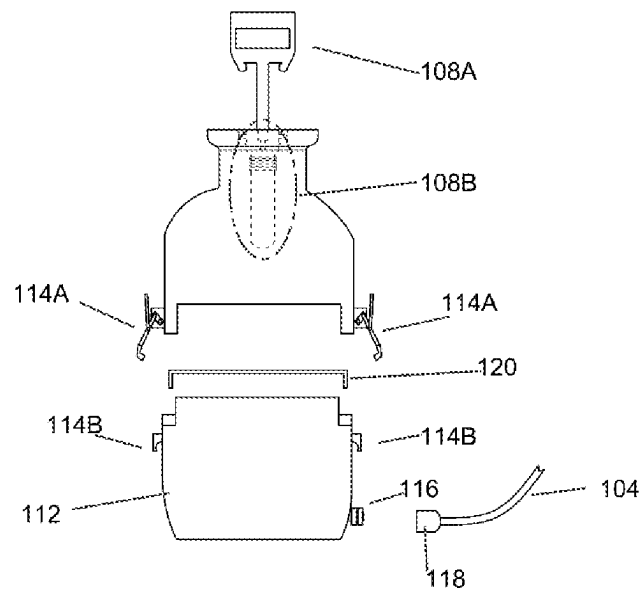


FIG. 2B

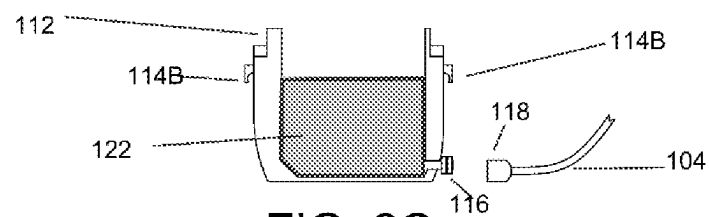
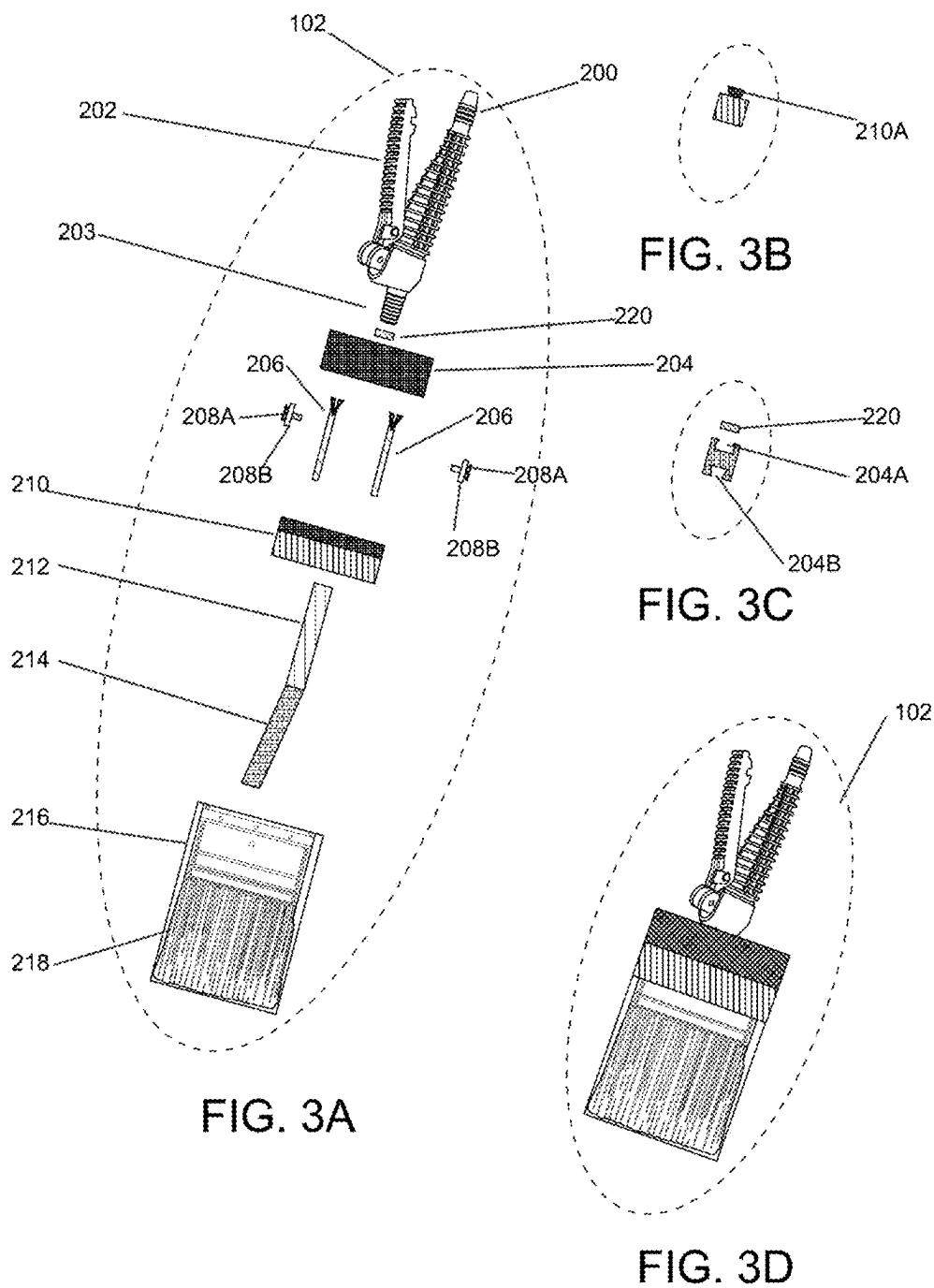


FIG. 2C



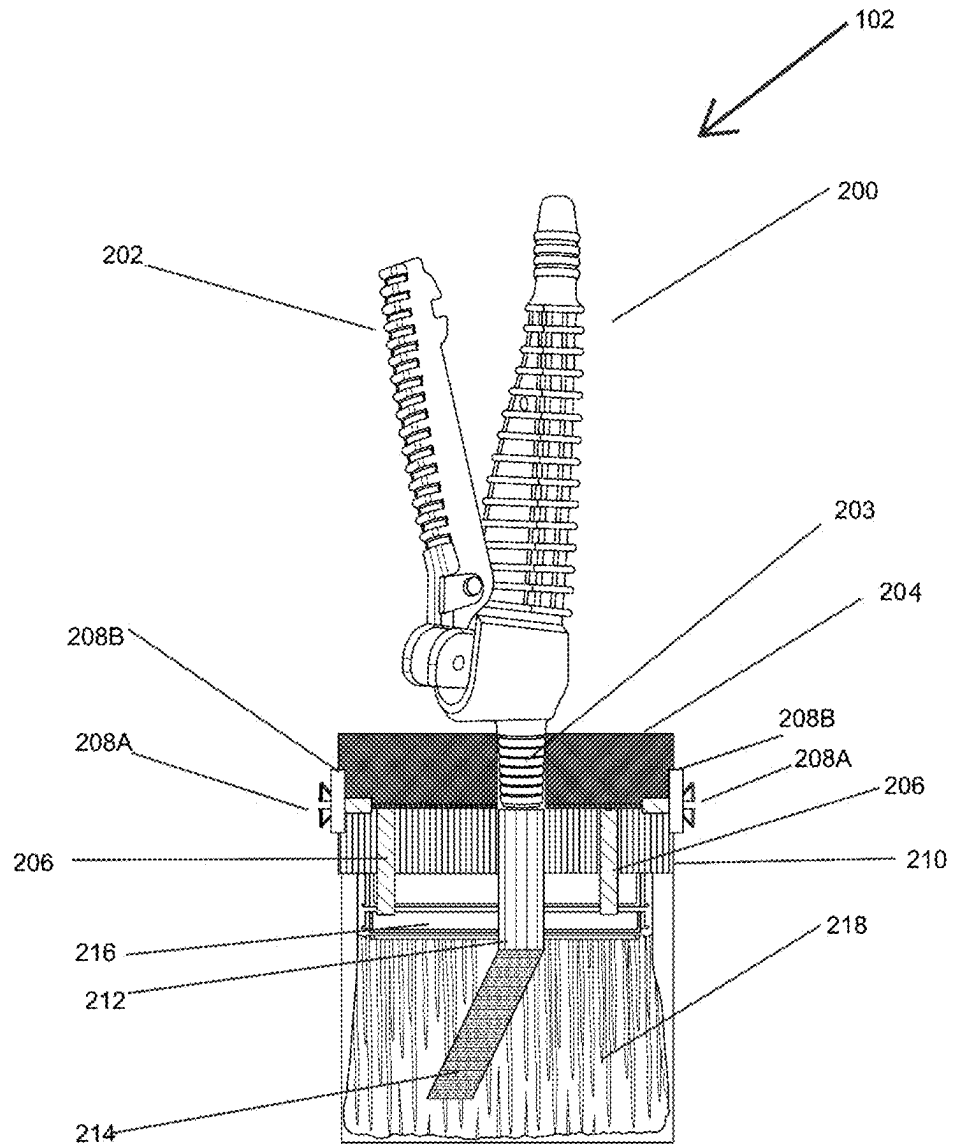


FIG. 4

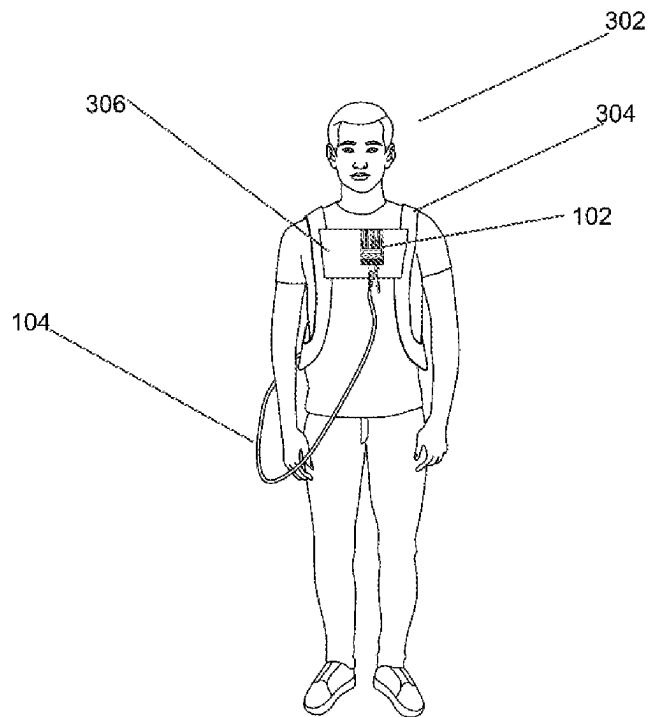


FIG. 5A

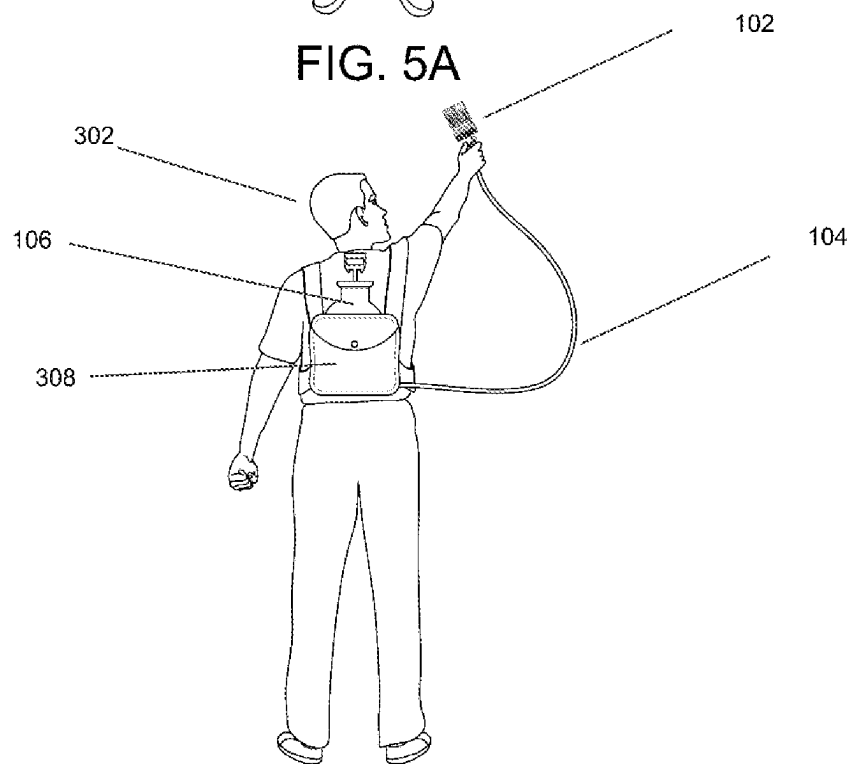


FIG. 5B

PORTABLE LIQUID COVER SYSTEM**BACKGROUND OF THE INVENTION**

The present disclosure relates to a novel device for 5
applying paint or other liquid covering material onto a
surface via a brush.

SUMMARY OF THE INVENTION

Broadly, the invention is a portable liquid cover system,
such as for painting, providing mobility and safety to the
person using it. The invention can be used to apply a variety
of liquid cover materials to surfaces, such as varnish or stain
for finishing wood, ink for large scaled calligraphy or
drawing such as artwork for murals, or adhesives. It can also
be used in various industrial applications, such as coating
metals or plastics with lubricants or other liquids.

The invention includes a hand air pump screwed into the
top portion of a vessel. The top part is cinched with the lower 20
paint chamber, which has a rubber bladder to keep the air
and paint separated. Pumping the air pump pressurizes the
rubber bladder, forcing the paint in the lower paint chamber
through a tube at the bottom of the chamber to the hand
valve.

The hand valve has a machined and a molded end that can
accommodate paintbrushes of different sizes by sliding them
together. Once attached, a screw or washer secures the brush
in place. When the hand valve lever is depressed, it regulates
the amount of paint flowing to the bristles through a pliable,
flexible tube, which moves with the brush in any direction.
This pliable tube can be made at different lengths along the
brush bristles to cater to various applications, be it painting
walls or ceilings. The vessel, along with the hose and
paintbrush, can be carried in a backpack. The backpack 35
features a zipper area or slot to accommodate the hose
connection. A magnet or spring-loaded clip across a front
panel of the backpack, in the chest area, secures the paint-
brush in an upright position to prevent dripping. This
provides significant safety for the person using this, allow-
ing the user to have both hands free to move about and climb
a ladder safely.

The materials used to make this invention include plastic
vessel, rubber air pump, plastic hand valve, machined alu-
minum for sliding together to change the brush, steel tubing,
wooden paintbrush, rubber balloon, and screws.

**DETAILED DESCRIPTION OF THE
INVENTION**

In the following detailed description, reference is made to
the accompanying drawings, which form a part hereof, and
in which are shown various specific embodiments in which
the disclosure may be practiced. These embodiments are
described in sufficient detail to enable those skilled in the art
to practice the disclosure, and it is to be understood that
other embodiments may be utilized and that mechanical,
procedural, and other changes may be made without depart-
ing from the spirit and scope of the present disclosure. The
following detailed description, therefore, is not to be taken
in a limiting sense. The scope of the present disclosure is
defined only by the appended claims, along with the full
scope of equivalents to which such claims are entitled.

The following reference numbers are used in the figures
and accompanying descriptions:

The following reference numbers are used in the figures
and accompanying descriptions:

No.	Name
100	liquid cover system
102	brush
104	hose
106	vessel
108A	pump handle
108B	pump body
110	top chamber
112	bottom chamber
114	latch
114A	top latch
114B	bottom latch
116	outlet
118	connector
120	diaphragm
122	liquid cover
200	brush handle
202	lever
203	outlet
204	top joint
204A	input hole
204B	groove
206	screw
208A	set screw
208B	washer
210	bottom joint
210A	tongue
212	rigid tube
214	pliable tube
216	binder
218	bristles
220	"O" ring
302	person
304	backpack
306	front panel
308	back compartment

BRIEF DESCRIPTION OF DRAWINGS

For a fuller understanding of the nature and advantages of
the present invention, reference should be had to the fol-
lowing detailed description taken in connection with the
accompanying drawings, in which:

The figures here represent one embodiment of the inven-
tion.

FIG. 1 is a view of the completed liquid cover system,
comprising a vessel, hose, and brush.

FIG. 2A shows various important components of the
assembled vessel.

FIG. 2B is an exploded view, principally showing cham-
bers and a diaphragm between them.

FIG. 2C is an interior view of the bottom chamber.

FIG. 3A is an exploded view of the brush.

FIGS. 3B and 3C are side views, respectively, bottom
joint and top joint.

FIG. 3D shows the assemble brush.

FIG. 4 is a cut away view of the assembled brush.

FIGS. 5A and 5B show a person using the liquid cover
system with a backpack.

DETAILED DESCRIPTION OF THE DRAWINGS

In the following detailed description, reference is made to
the accompanying drawings, which form a part hereof, and
in which are shown various specific embodiments in which
the disclosure may be practiced. These embodiments are
described in sufficient detail to enable those skilled in the art
to practice the disclosure, and it is to be understood that
other embodiments may be utilized and that mechanical,
procedural, and other changes may be made without depart-
ing from the spirit and scope of the present disclosure. The

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following detailed description, therefore, is not to be taken in a limiting sense. The scope of the present disclosure is defined only by the appended claims, along with the full scope of equivalents to which such claims are entitled.

FIG. 1 shows a liquid cover system 100 as assembled, including a vessel 106, a brush 102, and a hose 104 connected to the vessel 100.

FIG. 2A identifies key parts of the liquid cover system, including vessel 106 having an upper chamber 110, lower chamber 112, latches 114 that hold the upper 110 and lower 112 chambers together, outlet 116, hose 104 delivering liquid cover to the brush, and connector 118. Pump handle 108A is used in conjunction with pump body 108B (discussed below) to create air pressure in the upper chamber. The shape of pump handle 108A and its attachment to pump body 108B show that the pump handle 108A is arm operated, by person 302 (discussed below with respect to FIG. 5A), as would any pump with a handle similar to pump handle 108A.

FIG. 2B is an exploded view. Diaphragm 120 is deployed between upper chamber 110 and lower chamber 112, to separate liquid cover in the lower chamber from air in the upper chamber in an air-tight manner. The pump handle 108A is pumped to cause pump body 108B to create air pressure in the upper chamber 110. The diaphragm 120 is flexible so that air pressure applied to the upper chamber applies corresponding pressure, without leakage, in the lower chamber to force liquid cover through outlet 116 and through the hose 104, via secure connection provided by connector 118 attached to outlet 116. Furthermore, the diaphragm 120 may be disposable so as not need to be cleaned, or of a more permanent design which would require cleaning.

FIG. 2C is an interior view of the bottom chamber, principally showing liquid cover 122.

FIG. 3A is an exploded view of brush 102. Brush handle 200 includes a lever 202 to control paint flow, and a threaded outlet 203. Brush handle 200 is screwed into top joint 204 and hole input 204A, with O ring 220 between them. Rigid tube 212 and pliable tube 214 connected to it are inserted into bottom joint 210. Then the combination of bottom joint 210, rigid tube 212, and pliable tube 214 are inserted brush binder 216 and secured by screws 206, so that rigid tube 212 remains straight along the brush axis while pliable tube 214 departs from the rigid tube at an angle toward a leading edge of the bristles 218, the outlet of the pliable tube being positioned closer to said leading edge than to an opposite edge of the bristles. This configuration delivers liquid cover 122 to the leading edge, allowing trailing bristles to spread the coating evenly.

Refer now to FIGS. 3B and 3C for side views, respectively, of bottom joint 210 and top joint 204, showing the sliding dovetail attribute with groove 204B in top joint 204 and tongue 210A in bottom joint 210. Top joint 204 and bottom joint 210 are slid into place and alignment, each with holes reaching through to deliver liquid cover 122 through brush handle 200 and eventually through pliable tube 214 and into bristles 218. This connection between top joint 204 and bottom joint 210 is secured by a pair of set screws 208A and washers 208B.

FIG. 3D shows the final, assembled brush 102.

For clarification, FIG. 4 is a cut away view of assembled brush 102, further showing placement and connection of the various components.

FIGS. 5A and 5B introduce a person 302 wearing a backpack 304. The backpack 304 includes front panel 306 and back compartment 308. Back compartment 308 is

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designed to hold vessel 106, and to allow hose 104 to be connected to outlet 116. Front panel is provided with some means, such as a clip or magnet, to hold brush 102 while not being actively used to apply liquid cover 122 to a surface.

In addition to the advantages of this invention listed above, several ergonomic and efficiency advantages and benefits accrue in a liquid cover system 100 that does not require a person 302 to bend over to get paint on a brush 102. These include reduced strain on the back. By providing an alternative method for getting paint onto the brush 102, the person 302 can maintain a more neutral and comfortable position, reducing the risk of back pain and injury associated with prolonged bending.

Another benefit is improved posture. With very limited bending over, a person 302 can maintain a more upright and neutral posture, which can reduce stress on the back and other parts of the body. By keeping the body in a more ergonomic position, the person 302 can help prevent issues such as muscle tension, discomfort, and fatigue.

Yet another benefit is increased efficiency. By limiting the need to bend over, a person 302 can work more efficiently and with less fatigue. This can help reduce the risk of mistakes, risk and danger of spillage, and increase productivity.

Still further, another benefit is better access. The liquid cover system 100 keeps the paint or other liquid cover 122 readily available to apply to a surface, and at reduced risk of spillage. Person 302 can more easily access the liquid cover 122 without having to reach, stretch, or strain. This also reduces the risk of spillage. All of this can help to reduce the risk of slips, trips, and falls, as well as improve overall safety in the workspace.

Although the invention has been described with reference to various embodiments, those skilled in the art will understand that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope and essence of the disclosure. In addition, many modifications may be made to adapt a particular situation or material in accordance with the teachings of the disclosure without departing from the essential scope thereof. Therefore, it is intended that the disclosure not be limited to the particular embodiments disclosed, but that the disclosure will include all embodiments falling within the scope of the appended claims. Also, any and all citations referred to herein are expressly incorporated herein by reference.

I claim:

1. A system for applying liquid cover to a surface, comprising

a vessel (106) having an upper chamber (110) and a lower chamber (112) separated by a flexible diaphragm (120), wherein said flexible diaphragm (120) is disposable, latches (114) holding the upper chamber (110) and lower chamber (112) together,

an outlet (116) for liquid cover (122),

a hose (104) connected to the outlet (116),

a brush (102) adapted to receive liquid cover (122) from the hose (104) via a pliable tube (214), the brush (102) further having bristles (218),

the brush (102) further having a brush handle (200) and a lever (202), wherein the lever (202) is manually operable,

a pump handle (108A) wherein the pump handle (108A) is connected to a pump body (108B), further wherein the upper chamber (110) is pressurized by operation of the pump handle (108A) and the pump body (108B), wherein the pump handle (108a) is arranged to receive

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an applied force, which causes pressure in the upper chamber (110) to apply corresponding pressure to the lower chamber (112) to force the liquid cover (122) through the outlet (116) and hose (104) to the brush (102).

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2. The system of claim 1, wherein the brush (102) further comprises a rigid tube (212) attached to the pliable tube (214), the rigid tube extending along a longitudinal axis of the brush handle and the pliable tube extending from the rigid tube at an angle toward one edge of the bristles (218) 10 such that an outlet end of the pliable tube is nearer to said edge than to an opposite edge of the bristles, thereby delivering liquid cover (122) to that edge for efficient spreading by trailing bristles.

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